

RAJ SINGH

+91 7048960043 | rajasm029@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Experience

Full - Stack Engineering Intern

July 2026 – Present

Lowfied.com (Music Marketplace)

- Built core product systems for a pre-launch music marketplace with a **300+ user waitlist**, using **Next.js**, **NestJS**, and **PostgreSQL**.
- Engineered **Razorpay payment infrastructure**, implementing gateway abstractions, payment processing, webhooks, order state management, and subscription activation.
- Redesigned core modules around **domain-driven, layered architecture**, migrating **79+ API routes** across authentication, catalog, finance, licensing, and community domains.
- Optimized catalog architecture and caching, achieving a reported **4× improvement in load time** while migrating legacy modules to the new architecture.

AI Engineering Intern

January 2026 – June 2026

Ravan.ai (Enterprise Vocal AI ecosystem)

- Built production features across **3 AI-calling products**—CloserX, Snowie.ai, and Agni.ai—using **Next.js**, React, Zustand, and TanStack Query.
- Built an end-to-end **AI video ad generation platform** integrating **Claude Opus**, **Kling**, and **Gemini** to generate **10–20 second ads** from product information, characters, and voice selections.
- Designed the generation backend with **Django**, **PostgreSQL**, and **BullMQ**, orchestrating long-running AI jobs with polling, retries, reprocessing, partial-failure recovery, and AI credit accounting.
- Diagnosed and fixed Linux-specific **WebGPU** video preview issues in **TooCut**, contributing the fix upstream.

CyberSec Internship

June 2025 – August 2025

Software Technology Parks of India (An autonomous body under MeitY)

- Conducted vulnerability assessments across **50+ websites** using Nmap, Wireshark, and Burp Suite; mapped findings to **OWASP Top 10** and CVE/CWE IDs.

Projects

Web to Anki — Flashcard SaaS | *FastAPI, Python, JavaScript, PostgreSQL, LLM APIs*

- Built an end-to-end **AI-powered SaaS** tool — LLM pipeline extracts and summarizes web content into Anki flashcards, accelerating study by **10×**.
- Engineered a **Manifest V3** browser extension using Service Workers and async API calls to handle **1,000+ weekly requests** with optimized background performance.
- Monetized via PayPal and Ko-fi; drove a marketing campaign achieving **50k+ impressions** and **16 recurring paid users**.

SmartScape: Urban Infrastructure Planning Assistant | *Flask, Vertex AI, GCP* | IBM / CSRBOX

- Ranked **Top 4 pan-India** at the IBM SkillsBuild Program; showcased at IBM Summit 2025 and **AI Impact Global Summit 2026**.
- Engineered a full-stack **geospatial analytics** platform leveraging **Google Gemini multimodal AI** for urban planning insights.
- Deployed on **Render** with a Git-based CI/CD pipeline; secured Google Maps API via domain restriction.

Skills

- Languages:** TypeScript, Python, Java, SQL, Bash
- Backend:** NestJS, Django, FastAPI, PostgreSQL, Prisma, SQLAlchemy
- Frontend:** Next.js, React, Tailwind CSS, Zustand, TanStack Query
- AI / Systems:** LLM APIs, AI Agents, RAG, BullMQ, Redis
- Infrastructure:** Docker, Linux, Git, CI/CD

Education

Vivekananda Institute of Professional Studies
B.Tech in Computer Science and Engineering

Pitampura, Delhi
June 2026